be 1350 hours with a standard deviation of 100 hours. Establish 90% confidence limits within which the mean life time of light bulbs is expected to lie.

Solution:

Given: n = 169, $\bar{x}$ = 1350 hours, σ = 100 hours, since the level of significance is (100-90)% =10% thus α is 0.1, hence the significant value at 10% is $Z_{\alpha/2}$ = 1.645

$$S.E. = \frac{\sigma}{\sqrt{n}} = \frac{100}{\sqrt{169}} = 7.69$$

Hence 90% confidence limits for the population mean are

$$\bar{x} - Z_{\alpha/2} SE < \mu < \bar{x} + Z_{\alpha/2} SE$$

$$1350 - (1.645 \times 7.69) \leq \mu \leq 1350 + (1.645 \text{ x } 7.69)$$

$$1337.35 \leq \mu \leq 1362.65$$

Hence the mean life time of light bulbs is expected to lie between the interval (1337.35, 1362.65)

8.3 Hypothesis Testing

One of the important areas of statistical analysis is testing of hypothesis. Often, in real life situations we require to take decisions about the population on the basis of sample information. Hypothesis testing is also referred to as "Statistical Decision Making". It employs statistical techniques to arrive at decisions in certain situations where there is an element of uncertainty on the basis of sample, whose size is fixed in advance. So statistics helps us in arriving at the criterion for such decision is known as Testing of hypothesis which was initiated by J. Neyman and E.S. Pearson.

For Example: We may like to decide on the basis of sample data whether a new vaccine is effective in curing cold, whether a new training methodology is better than the existing one, whether the new fertilizer is more productive than the earlier one and so on.

8.3.1 Meaning : Null Hypothesis and Alternative Hypothesis - Level of Significants and Type of Errors

Statistical Hypothesis

Statistical hypothesis is some assumption or statement, which may or may not be true, about a population.

There are two types of statistical hypothesis

(i) Null hypothesis (ii) Alternative hypothesis

Null Hypothesis

Definition 8.7

According to Prof. R.A.Fisher, "Null hypothesis is the hypothesis which is tested for possible rejection under the assumption that it is true", and it is denoted by H_0.

For example: If we want to find the population mean has a specified value μ_0, then the null hypothesis H_0 is set as follows $H_0 : \mu = \mu_0$

Alternative Hypothesis

Any hypothesis which is complementary to the null hypothesis is called as the alternative hypothesis and is usually denoted by H_1.

For example: If we want to test the null hypothesis that the population has specified mean μ i.e., $H_0 : \mu = \mu_0$ then the alternative hypothesis could be any one among the following:

(i) $H_1 : \mu \neq \mu_0$ ($\mu >$ or $\mu < \mu_0$)

(ii) $H_1 : \mu > \mu_0$

(iii) $H_1 : \mu < \mu_0$

The alternative hypothesis in $H_1 : \mu \neq \mu_0$ is known as two tailed alternative test. Two tailed test is one where the hypothesis about the population parameter is rejected for the value of sample statistic falling into either tails of the sampling distribution. When the hypothesis about the population parameter is rejected only for the value of sample statistic falling into one

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